

DK LEARNING

GRADE 7

Algorithms & Python

Level: **Advanced** • Purpose: Learn before you practise

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In this guide

1. Algorithms
2. Python Basics: print and input
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4. Decisions with if / else
5. Loops
6. Practice questions, answers and glossary



How to use this guide

Read each chapter slowly. Look at the example, then try the “Your turn” task **without copying**. Write your own notes on the lines. When you finish, answer the practice questions and check your work with the answer page. After that, open the Grade 7 activity on the website.

Learning goals

- Write an algorithm in plain steps and flowchart ideas
- Use variables and data types in Python
- Make decisions with if / else
- Repeat actions with loops

Study plan

Step	What to do
1	Read the chapter and highlight key words
2	Study the worked example
3	Do the “Your turn” task on paper or on a computer
4	Write a short summary in your own words
5	Answer the practice questions, then check

Tip: Run every code example yourself and then change a value. Experimenting is how programmers learn.

1 Algorithms

An algorithm is a clear sequence of steps used to solve a problem. Algorithms can be written in plain language or pseudocode before coding. Good algorithms are clear, finite and give the correct result.

Flowcharts show an algorithm with shapes: oval for start/end, rectangle for a process, diamond for a decision.

Key points

- Clear ordered steps
- Must end after a finite number of steps
- Pseudocode is plain-language code
- Flowcharts use shapes and arrows

Worked example

Making tea:

1. Boil water
2. Put tea in the cup
3. Pour water
4. Wait 3 minutes
5. Add milk or sugar if you like

Your turn

Write an algorithm for finding the larger of two numbers.

Check yourself

Explain each term in your own words:

Algorithm

Pseudocode

My notes

2 Python Basics: print and input

Python is a popular programming language that is easy to read. `print()` shows output. `input()` asks the user to type something. Text values are called strings and are written in quotes.

Python reads code from top to bottom. Indentation (spaces at the start of a line) matters.

Key points

- `print()` shows output
- `input()` reads text from the user
- Strings are in quotes
- Python is case-sensitive

Worked example

```
print("Hello DK Learning")
name = input("Your name: ")
print("Welcome,", name)
```

Your turn

Write a program that asks for your name and favourite subject and prints both.

Check yourself

Explain each term in your own words:

String

Function

My notes

3 Variables and Data Types

A variable stores a value that can change. In Python, `score = 10` creates a variable named `score`. Common data types are `int` (whole numbers), `float` (decimal numbers), `str` (text) and `bool` (True or False).

Choose meaningful names such as `total_marks`, not `x1`. Use `=` to assign a value.

Key points

- Variable = named storage
- `int`, `float`, `str`, `bool`
- Use meaningful names
- `=` assigns a value

Worked example

```
score = 10
score = score + 5
print(score)      # 15
price = 12.5      # float
name = "Nimal"    # str
```

Your turn

Trace: `a = 4`; `b = 3`; `a = a + b`. What are `a` and `b`?

Check yourself

Explain each term in your own words:

Variable

Data type

My notes

4 Decisions with if / else

Selection lets a program make decisions with if / else. A condition is checked. If it is True, one block runs; otherwise another block runs. Comparison operators are ==, !=, <, >, <= and >=.

Remember the colon (:) after the condition and indent the block.

Key points

- if checks a condition
- else runs when False
- == compares; = assigns
- Indent the block

Worked example

```
mark = 62
if mark >= 50:
    print("Pass")
else:
    print("Try again")
```

Your turn

Write code that prints "Adult" if age is 18 or more, otherwise "Minor".

Check yourself

Explain each term in your own words:

Condition

Operator

My notes

5 Loops

Loops repeat instructions, reducing repeated code. A for loop repeats a set number of times. A while loop repeats while a condition stays True.

Be careful: if the condition never becomes False, a while loop runs forever.

Key points

- for = repeat a number of times
- while = repeat while True
- range(1, 6) gives 1 to 5
- Avoid infinite loops

Worked example

```
for i in range(1, 6):  
    print(i)  
  
total = 0  
while total < 3:  
    total = total + 1
```

Your turn

Write a loop that prints the 5 times table.

Check yourself

Explain each term in your own words:

Loop

Iteration

My notes



Practice Questions

Part A - Choose the best answer

1. An algorithm is...
 - A) a clear sequence of steps
 - B) a random guess
 - C) a keyboard
2. Which function shows output in Python?
 - A) print()
 - B) show()
 - C) out()
3. `score = 10` creates a...
 - A) variable
 - B) loop
 - C) picture
4. Which compares two values?
 - A) `==`
 - B) `=`
 - C) `:`
5. `range(1, 4)` gives...
 - A) 1,2,3
 - B) 1,2,3,4
 - C) 0,1,2,3
6. A loop that never stops is called...
 - A) infinite
 - B) short
 - C) empty

Part B - Short answers

1. What is pseudocode?

2. What is the output? `x = 2; x = x + 3; print(x)`

3. Difference between = and ==?

4. What does a for loop do?



Answer Page

Part A

1-A 2-A 3-A 4-A 5-A 6-A

Part B (sample answers)

1. Plain-language planning of code.
2. 5
3. = assigns a value; == compares two values.
4. Repeats instructions a set number of times.

Mini glossary

Algorithm	Step-by-step method to solve a problem
Pseudocode	Planning code written in simple language
String	Text data written in quotes
Function	A named block of code that does a job
Variable	A name that stores a value
Data type	The kind of value stored
Condition	A question that is True or False
Operator	A symbol that does an operation
Loop	Code that repeats
Iteration	One repeat of a loop



Activity Preparation

Solve algorithm questions and write a small Python solution using variables, decisions or loops. Then open the Grade 7 activity.

Think before you start

- Explain one important idea from this grade in your own words.
- Give one real-life example where this topic is useful.
- What mistake could a beginner make, and how can it be prevented?
- Write the steps you will follow before starting the Grade 7 activity.

Ready checklist

- | | |
|---|--|
| ☐ I read every chapter | ☐ I tried the “Your turn” tasks |
| ☐ I understand the key terms | ☐ I checked my practice answers |
| ☐ I wrote my own notes | ☐ I am ready for the activity |

⚡ **START GRADE 7 ACTIVITY**

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